Ecumenical Negation: One or Two?

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Abstract

Prawitz's ecumenical logic was designed to allow classical and intuitionistic reasoning to coexist within a single deductive framework. The resulting language contains both classical and intuitionistic versions of implication, disjunction and existential quantification, while retaining a single negation. This asymmetry raises a natural question. If negation is traditionally understood as implication into absurdity, why should the distinction between classical and intuitionistic implication not induce a corresponding distinction between classical and intuitionistic negation? We revisit this question from a proof-theoretic perspective. Interderivability, normalization, Glivenko-style arguments and computational isomorphism all suggest that intuitionistic and classical negation behave as a single connective. One might expect that once implication bifurcates into classical and intuitionistic forms, negation should bifurcate as well. Surprisingly, this does not happen. Rather than supporting a reduction of negation to implication into absurdity, the ecumenical setting suggests that negation enjoys a remarkable stability across the classical/intuitionistic divide. This raises the possibility that negation is more primitive, from a proof-theoretic perspective, than is often assumed.

1 Introduction

Ecumenical logic arose from an attractive philosophical idea due to Prawitz. Instead of asking whether classical logic or intuitionistic logic is the correct logic, one may seek a common framework in which both styles of reasoning can coexist.

The project is motivated by mathematical practice itself. Constructive and classical arguments often coexist within the same mathematical development. Rather than forcing a choice between foundational standpoints, ecumenical logic seeks a language that makes explicit where the two traditions agree and where they diverge.

Prawitz's ecumenical natural deduction system contains two implications, two disjunctions and two existential quantifiers, one intuitionistic and one classical. At the same time, it contains only one conjunction, one universal quantifier, one constant for absurdity and one negation.

The asymmetry is striking.

For the first author, the question was initially motivated by a different expectation. Negation is often treated as a derived connective, definable as implication into absurdity. From this perspective, once implication is duplicated into classical and intuitionistic forms, one naturally expects a corresponding duplication of negation. Several rounds of discussion with Luiz Carlos Pereira and Dag Prawitz gradually suggested that the ecumenical setting points in the opposite direction. The issue is not merely whether there are one or two negations, but whether negation should be regarded as a derived operation at all.

If implication comes in two flavours, and if negation is usually understood as implication into absurdity, why should there be only one negation? Why should we not distinguish between

$$\neg_i A := (A \to_i \bot)$$

and

$$\neg_c A := (A \to_c \bot)?$$

At first sight this may appear to be a purely technical question. We shall argue that it is better understood as a question about the identity of logical constants.

Recent work on Full Intuitionistic Logic[?] and Full Intuitionistic Linear Logic[?] suggests that traditional explanations of constructivity often focus on surface syntactic restrictions, while the deeper issue concerns the management of dependencies among assumptions and conclusions. From this viewpoint, ecumenical logic raises a related question: which distinctions between classical and intuitionistic connectives correspond to genuine structural differences and which do not? The case of negation in Ecumenical logic provides an illuminating test.

2 Prawitz's Ecumenical Logic

Prawitz proposed an ecumenical system [?] in which classical and intuitionistic reasoning coexist peacefully. Two rival logics coexist provided they validate the same theorems and preserve the same deductive relations.

The language contains

$$\wedge,\ \perp,\ \neg,\ \forall,$$

together with intuitionistic and classical versions of

$$\to,\ \vee,\ \exists.$$

Thus we have

$$\to_i,\ \to_c,\ \vee_i,\ \vee_c, \exists_i,\ \exists_c,$$

but only a single negation symbol.

The intuitionistic connectives are governed by the usual Gentzen–Prawitz introduction and elimination rules. Classical implication and classical disjunction are introduced through additional rules designed to capture classical reasoning while preserving normalization and other proof-theoretic properties.

The central design decision for the present paper is the presence of a unique negation operator.

3 Why Two Negations Seem Natural

Negation is traditionally explained as implication into absurdity.

In a setting with two implications, one is naturally led to define

$$\neg_i A := A \to_i \perp$$

and

$$\neg_c A := A \to_c \perp.$$

The question then becomes immediate: why should these two constructions be identified?

One possible answer is simply that they are interderivable.

Proposition 1. *The formulas*

$$(A \to_i \perp)$$

and

$$(A \to_c \perp)$$

are mutually derivable in the ecumenical system.

The proof follows directly from the introduction and elimination rules for the two implications.

At first sight, this seems decisive.

However, logical equivalence is often a weak notion of identity. Classically equivalent formulas need not have the same computational content, proof-theoretic behaviour, or explanatory role. Thus interderivability alone does not settle the issue.

4 Negation as an Assertability Condition

A second answer focuses on proof-theoretic meaning. In both intuitionistic and classical natural deduction, a normal proof of a negation ends with an application of negation introduction. To establish $\neg A$, one assumes A and derives absurdity. The assertability condition for negation therefore appears to be unique.

> To assert $\neg A$ is to derive a contradiction from the assumption A.

The resources employed inside the derivation may differ, but the form of the assertion remains unchanged. From this perspective, negation has a unique proof-theoretic meaning.

A natural objection remains. Even if the assertability condition is unique, perhaps there exist essentially classical derivations of contradiction. If so, these might justify a genuinely classical notion of negation.

5 Glivenko and the Absence of Classical Contradictions

In propositional logic, the preceding objection can be answered using Glivenko's theorem.

Suppose

$$A \vdash_c \bot.$$

Then

$$\vdash_c \neg A.$$

By Glivenko's theorem,

$$\vdash_i \neg A.$$

Hence

$$A \vdash_i \bot.$$

Thus every classical derivation of contradiction from a single assumption yields an intuitionistic derivation of contradiction from the same assumption.

Contradictions do not come in distinct classical and intuitionistic varieties, $\bot_c$ and $\bot_i$. (Should they? They do in linear logic, where we have $\bot$ and 0.)

This result already suggests that negation behaves differently from implication and disjunction.

6 Normalization and Seldin's Strategy

Additional evidence comes from normalization. Seldin showed that every classical derivation can be transformed into a derivation containing at most one use of the classical absurdity rule and that this application, when present, occurs as the final inference. When the conclusion is a negation, the final classical step may be eliminated entirely.

The resulting derivation is intuitionistic.

Consequently, proofs of negated propositions admit a normalization behaviour that does not depend on a specifically classical form of negation. Again the distinction appears to collapse.

7 Computational Isomorphism

The previous arguments rely on derivability and normalization. A stronger notion of equivalence is provided by computational isomorphisms.

Following Joinet, Danos and Schellinx[?], two propositions are computationally isomorphic when they admit mutually inverse proof terms.

Theorem 1.

$$(A \to_i \bot) \cong (A \to_c \bot).$$

The theorem establishes more than logical equivalence.

The two forms of negation possess the same computational content. Under the Curry–Howard correspondence they represent the same type.

This provides a considerably stronger justification for retaining a single negation in the ecumenical language.

8 Discussion

The original question of this paper was simple:

> Should ecumenical logic contain one negation or two?

The proof-theoretic evidence consistently points toward a single negation.

Interderivability, normalization, Glivenko's theorems, and computational isomorphisms all suggest that intuitionistic and classical negation fail to determine genuinely different logical operations. The more interesting conclusion is therefore negative.

If the distinction between classical and intuitionistic reasoning is not located in negation, then where is it located?

The evidence assembled in the previous sections consistently suggests that negation behaves differently from implication. Indeed, the original motivation for the investigation was the expectation that negation could simply be analysed as implication into absurdity. The ecumenical setting provides a natural test of that expectation. If negation is merely a definitional abbreviation for implication into absurdity, then one would expect the duplication of implication to induce a duplication of negation.

The surprising outcome is that this does not occur. The distinction between intuitionistic and classical implication survives, while the corresponding distinction between intuitionistic and classical negation repeatedly collapses. The phenomenon appears at the level of derivability, normalization, and computational content. Rather than reducing negation to implication, the ecumenical analysis suggests that negation possesses a proof-theoretic stability of its own.

9 Conclusions

The question of ecumenical negation began with a simple expectation. If negation is understood as implication into absurdity, and if ecumenical logic distinguishes classical and intuitionistic implications, then one might expect corresponding classical and intuitionistic negations.

The purpose of this paper was to examine that expectation.

The result is unexpectedly robust. Interderivability, normalization, Glivenko-style arguments and computational isomorphism all point in the same direction. Although implication splits into distinct intuitionistic and classical forms, negation does not. The distinction repeatedly disappears.

For us, the significance of this observation is not merely that ecumenical logic contains one negation instead of two. Rather, it casts doubt on a common way of thinking about negation itself. Throughout much of proof theory and logic, negation is presented as a derived connective, a convenient abbreviation for implication into absurdity. The ecumenical setting suggests a different picture. Negation appears considerably more stable than implication across the classical/intuitionistic divide.

We do not claim that implication should be explained in terms of negation. But the present analysis suggests that negation cannot be dismissed as a mere macro. Its proof-theoretic identity survives a distinction that affects implication in a fundamental way.

This observation raises broader questions about the identity of logical constants and about which proof-theoretic operations should be regarded as primitive. The broader lesson of this note is methodological. Counting connectives is less informative than understanding the structural roles they play. The challenge for ecumenical logic is not deciding how many logical constants should appear in the language, but identifying which distinctions correspond to genuine differences in proof, computation, and dependency. In this sense, the question "one negation or two?" opens onto a more general investigation of the architecture of logical systems. We hope that ecumenical logic will continue to provide a useful setting in which to investigate such questions.

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